Experiment : 3

Dimensionality Reduction Using Principal Component Analysis (PCA)

Mapped CO(s) : CO2, CO4

Aim

To perform dimensionality reduction using Principal Component Analysis (PCA) on the Titanic dataset using Python.

Objectives

- To understand the concept of dimensionality reduction.
- To preprocess data before applying PCA.
- To standardize numerical features.
- To apply PCA using Scikit-learn.
- To visualize the principal components.
- To analyze the explained variance of each principal component.

Prerequisites

- Basic knowledge of Python programming.
- Basic understanding of linear algebra and statistics.
- Python 3.x installed.
- Jupyter Notebook or Google Colab.
- Required libraries: Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

Theory/Background

Principal Component Analysis (PCA) is an unsupervised machine learning technique used for dimensionality reduction. It transforms a large set of correlated variables into a smaller set of new variables called **Principal Components**, while preserving as much information (variance) as possible.

The first principal component captures the maximum variance in the data. Each subsequent component captures the remaining variance while being orthogonal to the previous components.

PCA helps in:

- Reducing the number of features.
- Removing redundancy among correlated variables.
- Improving computational efficiency.
- Visualizing high-dimensional datasets.
- Preparing data for machine learning models.

Before applying PCA, numerical features are usually standardized so that each feature contributes equally to the analysis.

Algorithm/Methodology

- Import the required libraries.
- Load the Titanic dataset.
- Select numerical features.
- Handle missing values.
- Standardize the data using StandardScaler.
- Apply PCA to reduce dimensions.
- Calculate the explained variance ratio.
- Visualize the first two principal components.
- Interpret the results.

Dataset Description

Dataset Name: Titanic Dataset

Source: Seaborn / Kaggle

The Titanic dataset contains demographic and travel information of passengers aboard the RMS Titanic. In this experiment, only numerical features are used for PCA.

Numerical Attributes Used

Attribute	Description
age	Age of passenger
fare	Ticket fare
sibsp	Number of siblings/spouses aboard
parch	Number of parents/children aboard
pclass	Passenger class

Program/Implementation

The program should include:

- Importing required libraries.
- Loading the Titanic dataset.
- Selecting numerical features.
- Handling missing values.

- Standardizing data.
- Applying PCA.
- Displaying explained variance.
- Visualizing principal components using a scatter plot.

Test Cases

Test Case	Expected Result
Load Titanic dataset	Dataset loads successfully
Handle missing values	Missing values removed
Standardize features	Data scaled successfully
Apply PCA	Principal components generated
Plot PCA graph	Two-dimensional visualization displayed

Experimental Results

Observation

- Numerical features were successfully selected.
- Missing values were removed before PCA.
- Data was standardized using StandardScaler.
- PCA reduced the dataset to two principal components.
- Most of the information was retained after dimensionality reduction.
- The scatter plot clearly visualized the transformed data.

Result

Principal Component Analysis was successfully applied to the Titanic dataset, reducing the dimensionality while preserving most of the important information.

Conclusion

The experiment demonstrated how Principal Component Analysis (PCA) can effectively reduce the dimensionality of a dataset while retaining maximum variance. PCA simplifies data visualization, reduces computational complexity, and improves the efficiency of machine learning algorithms.

Exercises/Assignments

1. Apply PCA using three principal components.
2. Plot the explained variance ratio of each component.
3. Compare the dataset before and after PCA.

4. Interpret the first two principal components.
5. Apply PCA to another dataset.